Demographic Impact on Economic Growth

Analyzing the Impact of Demographic Factors, Including Population Growth, Age Structure, and Education, on Global Economic Growth Between 2013 and 2023

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ABSTRACT

This study examined the relationship between demographic factors and GDP growth from 2013 to 2023 using panel regression models and global economic data. Population growth emerged as a statistically significant driver of economic expansion in contexts with limited alternative growth factors (coefficient: 4.51, p = 0.004), while its effect diminished in more diversified economies. Other variables, including age structure and education levels, demonstrated limited explanatory power, with statistically insignificant results across all models. Low R² values (ranging from -0.25 to 0.10) indicate that demographic factors alone are insufficient to explain GDP variability, underscoring the importance of additional economic and institutional drivers. Refinements in the analysis improved model reliability, revealing the importance of robust data preparation and methodological rigor. The findings provide a nuanced understanding of the demographic-economic relationship, highlighting population growth as a key variable in less complex economies while suggesting further exploration of structural factors in advanced contexts.

1 INTRODUCTION

Background

Demographic factors, such as population growth, age structure, and education levels, play a pivotal role in shaping economic development. Over the past decade (2013–2023), global economies have faced challenges ranging from demographic shifts to economic disruptions caused by the COVID-19 pandemic. Understanding how these demographic trends influence economic outcomes is essential for policymakers and researchers striving to foster sustainable growth.

Research Questions

This study aims to answer the following primary research question:

How do demographic factors influence economic growth between 2013 and 2023?

To address this overarching question, three specific hypotheses are explored:

1. **Population Growth:** Does population growth drive economic growth?
2. **Age Structure:** How does age structure impact economic growth, and is an aging population negatively affecting it?
3. **Education Levels:** Does a more educated population correlate with higher economic growth?

Significance of the Study

By examining these questions, this research seeks to uncover the intricate relationships between demographic trends and economic outcomes. The findings aim to provide insights into the long-term economic implications of demographic changes and inform evidence-based policy decisions.

2 DATA

Data and Sources

To address the research questions, this study utilizes data from the World Bank's World Development Indicators (WDI), a comprehensive and widely recognized dataset. The WDI dataset offers annual, standardized measurements of demographic and economic variables, including population metrics, education levels, and GDP growth, across multiple countries. Data collection integrates inputs from national statistical offices, international organizations, and independent surveys, ensuring high reliability and accuracy.

The dataset spans a decade (2013–2023), allowing for a robust analysis of global trends. Key variables include:

Population Growth Rate: Annual percentage change in population size.

Age Groups: Proportions of populations aged 0–14, 15–64 (working age), and 65+.

Education Metrics: Enrollment rates and levels of educational attainment.

GDP Growth: Annual percentage growth in real GDP.

Time frame: The period from 2013 to 2023 provides a decade of data, offering a sufficient time frame to observe meaningful demographic and economic trends.

Countries: The report will focus on all countries available in the dataset, to capture global trends.

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Variable.Name	WDI.Code	Significance.to.the.Study
Population Growth	SP.POP.GROW	*Measures annual population growth, a key factor in assessing demographic changes.*
GDP Growth	NY.GDP.MKTP.KD.ZG	*Captures annual economic growth, providing insight into the relationship with demographics.*
Age 0–14	SP.POP.0014.TO.ZS	*Represents the percentage of the population aged 0–14, crucial for analyzing age dependency.*
Age 15–64	SP.POP.1564.TO.ZS	*Represents the working-age population, which impacts economic productivity and growth.*
Age 65+	SP.POP.65UP.TO.ZS	*Indicates the aging population, helping to assess its economic implications.*
Primary Enrollment	SE.PRM.ENRR	*Tracks primary education enrollment, linked to foundational human capital development.*
Secondary Enrollment	SE.SEC.ENRR	*Tracks secondary education enrollment, reflecting skill development for economic growth.*
Educational Attainment	SE.PRM.CUAT.ZS	*Measures the percentage of adults completing primary education, showing overall education levels.*
Income Level	income	*Indicates the income category of countries, providing context for economic and demographic analysis.*
Total Population	SP.POP.TOTL	*Measures the total population, essential for understanding demographic scale and trends.*
Real GDP	NY.GDP.MKTP.KD	*Reflects the real GDP, offering a measure of economic output adjusted for inflation.*

This combination of demographic and economic indicators enables a comprehensive exploration of the interplay between demographic shifts and economic performance.

3 EXPLORATORY DATA ANALYSIS

The table above presents summary statistics for the year 2015.

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Key Metrics for the Year 2015

	Value
Min_GDP_Growth	*-3.545763*
Median_GDP_Growth	*2.945550*
Mean_GDP_Growth	*3.316186*
Max_GDP_Growth	*24.615570*
Min_Pop_Growth	*-2.142484*
Median_Pop_Growth	*1.120993*
Mean_Pop_Growth	*1.215072*
Max_Pop_Growth	*8.581733*

Similar calculations have been performed for other years (2018 and 2022), but they are omitted here to maintain report brevity and readability.

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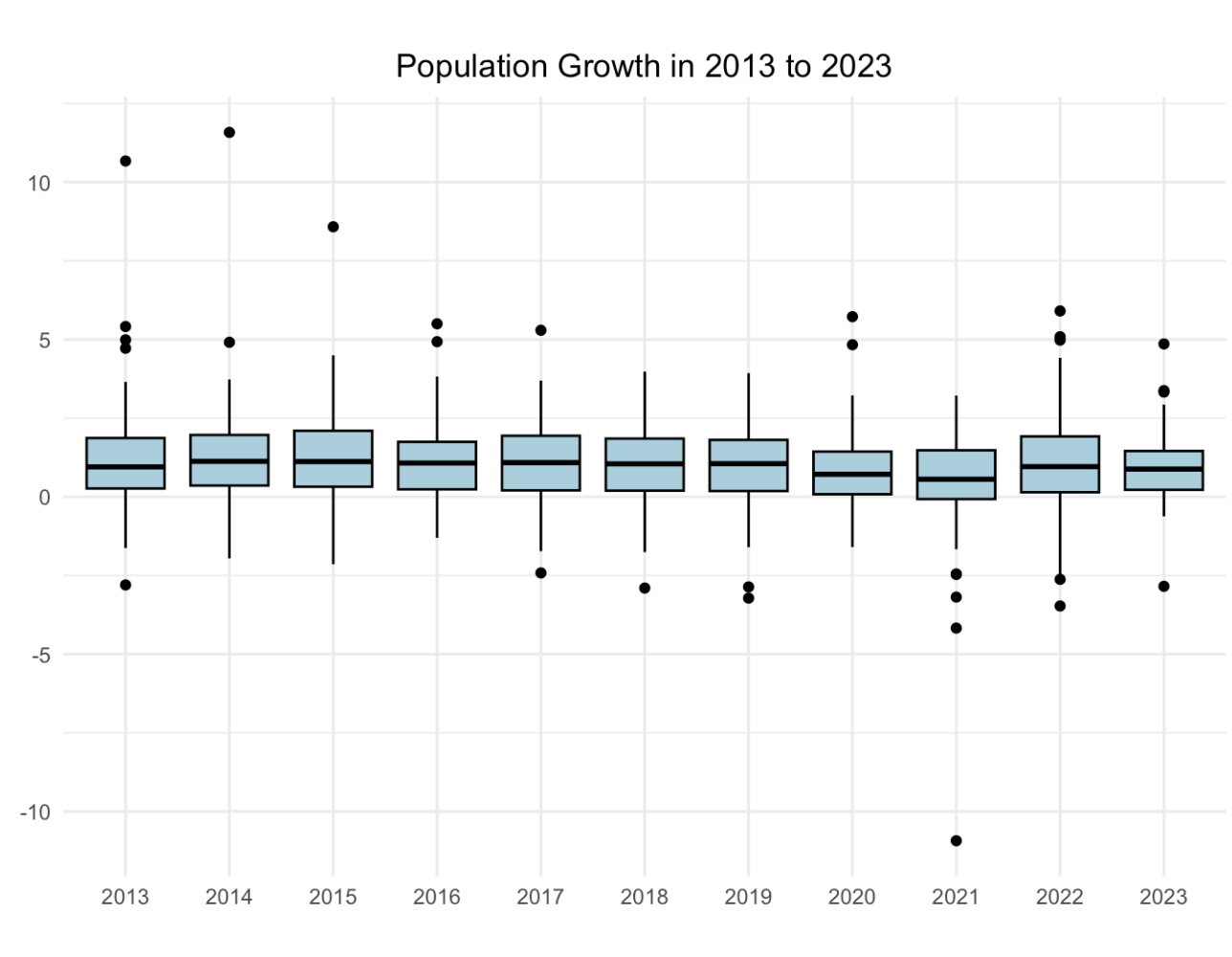

From 2013 to 2023, the global population growth rate exhibited a steady decline, beginning at 1.24% in 2013 and reaching 0.98% by 2020. This downward trend was further accelerated by the COVID-19 pandemic, which significantly impacted global population dynamics, reducing the growth rate to 0.83% in 2021. As the effects of the pandemic began to subside, a modest recovery was observed, with the growth rate rebounding slightly to 0.88% in 2023. This trajectory underscores the pandemic's immediate disruptive influence on demographic patterns, followed by a gradual stabilization in subsequent years.

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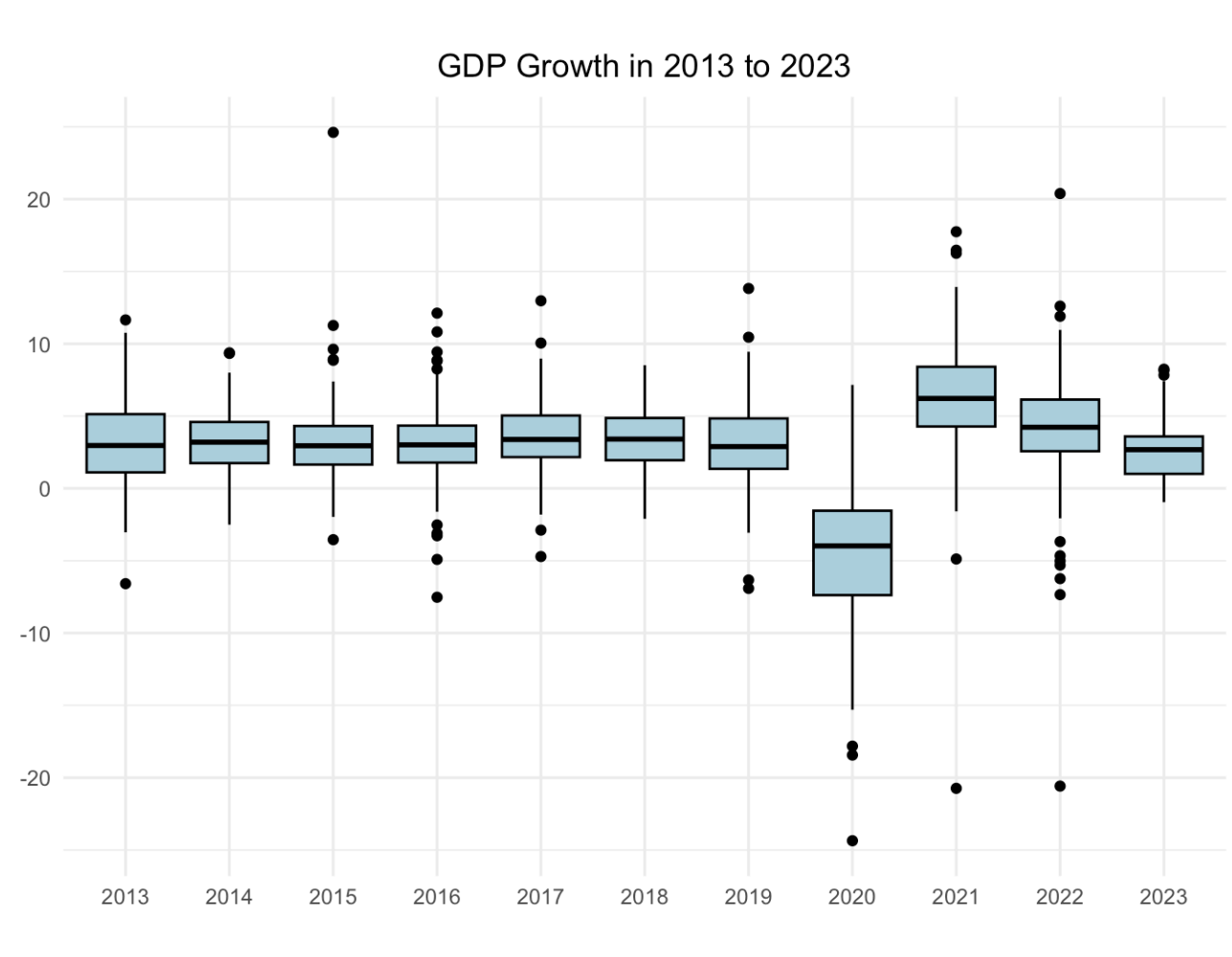

Over the past decade, global GDP growth demonstrated variability, generally oscillating between 2% and 3%. Starting at 2.87% in 2013, it peaked at 2.94% in 2019. However, the onset of the COVID-19 pandemic in 2020 triggered a severe contraction, with GDP growth plunging to -2.93% as lockdown measures and halted economic activity disrupted global markets. A swift rebound followed in 2021, with GDP growth surging to 6.26%, marking a period of recovery. Subsequently, growth stabilized within the 2%-3% range, reflecting the economic adjustments and lingering challenges of the post-pandemic environment.

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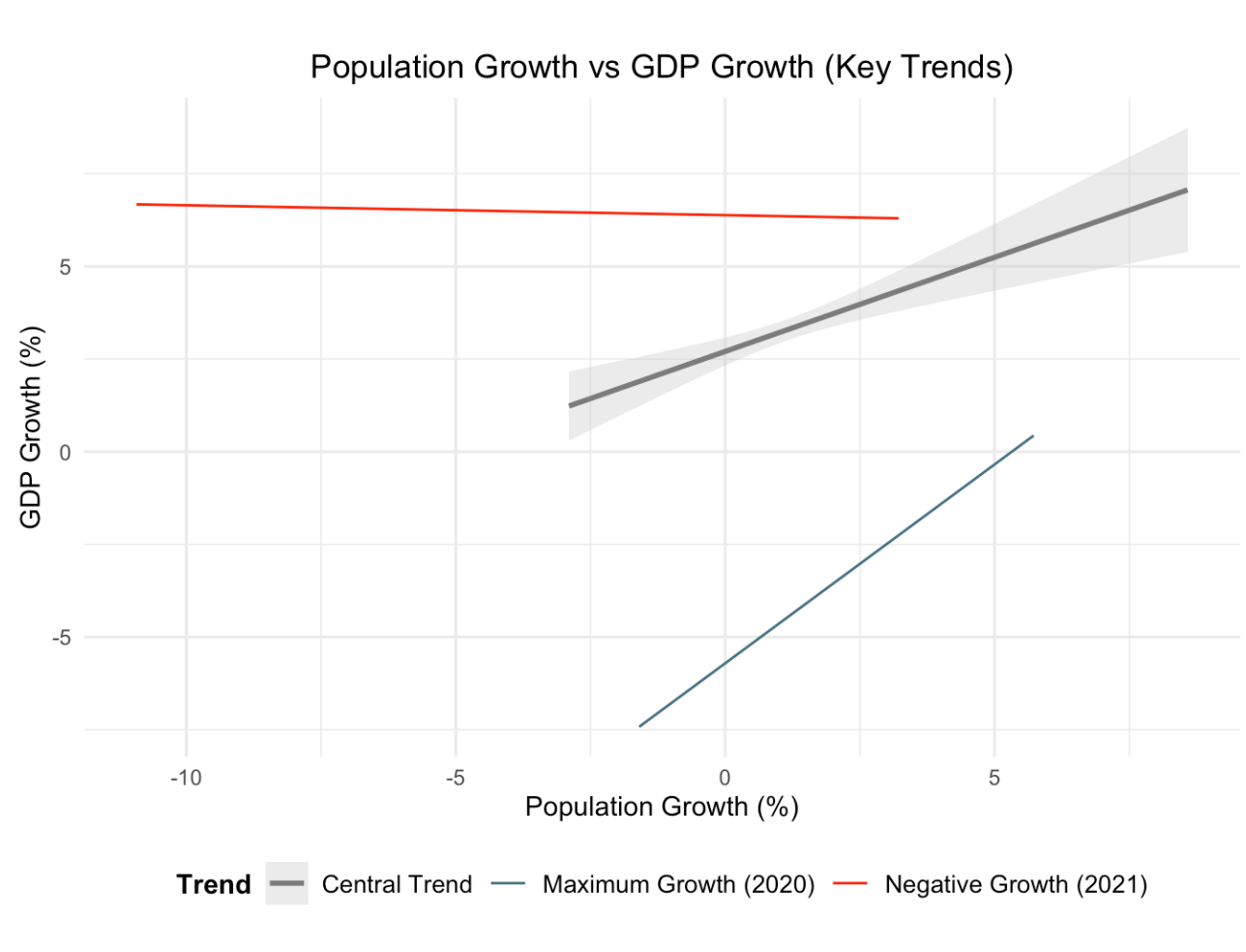

Positive regression coefficients observed across most years indicate a general positive relationship between population and GDP growth. Higher population growth often correlates with an expanding labor force, increased demand for goods and services, and stimulated economic production. Investments to support a growing population further boost economic activity.

In contrast, 2021 shows a negative regression coefficient, reflecting the COVID-19 pandemic's extraordinary impact. Widespread lockdowns, trade disruptions, and halted production decoupled GDP growth from population trends. While population growth remained steady, economic contraction in 2021 created a temporary negative correlation, driven by external health and economic crises.

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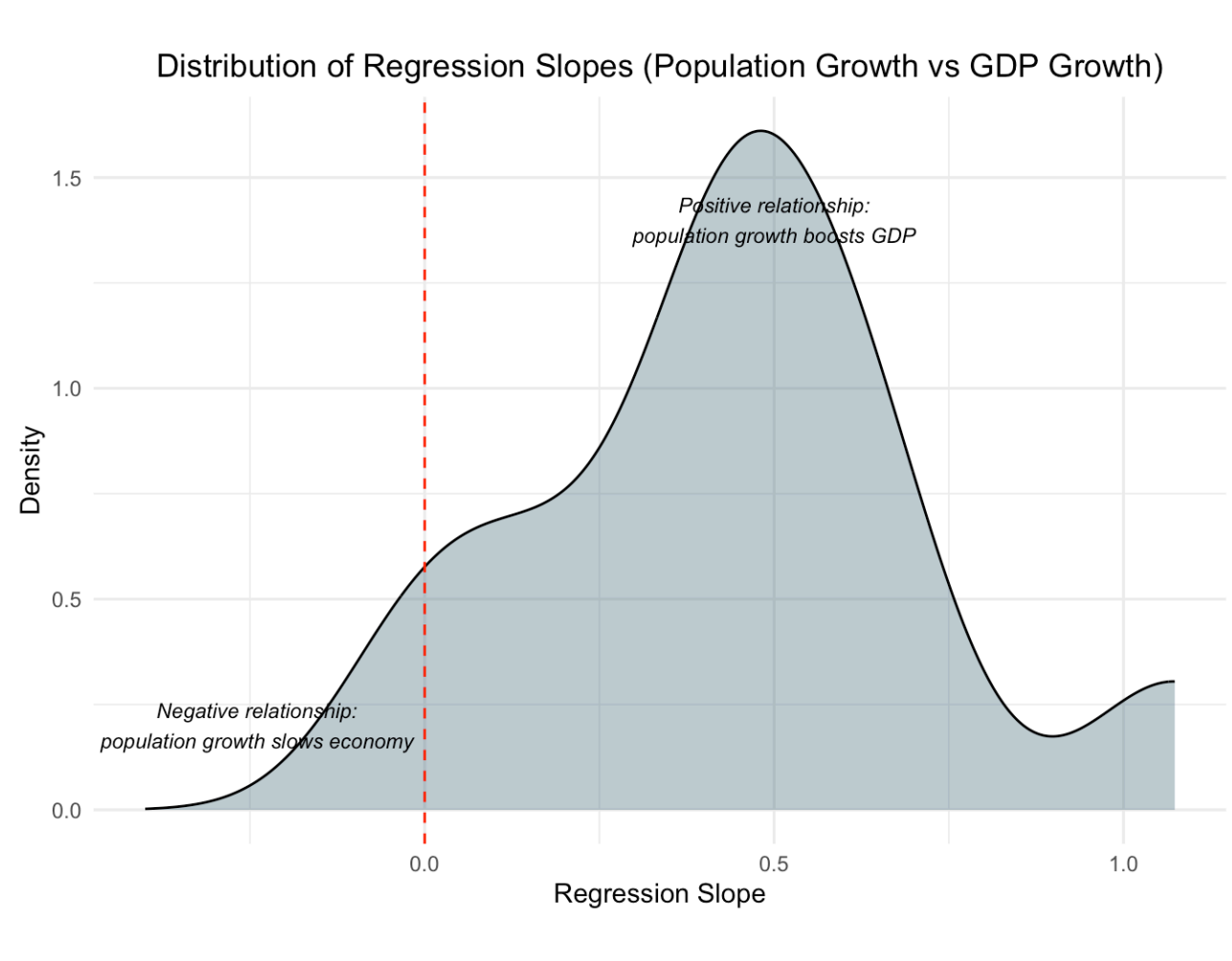

The graph illustrates the distribution of regression coefficients, representing the relationship between population growth and GDP growth over the study period. The peak around 0.3 indicates that population growth was most commonly associated with a moderate positive impact on the economy. The presence of negative values to the left of zero highlights rare instances where population growth slowed economic growth, likely due to external factors such as crises.

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Ranges for 1, 2, and 3 Sigma for Population and GDP Growth

Metric	X1.Sigma	X2.Sigma	X3.Sigma
Population Growth	-0.38 to 2.43	-1.78 to 3.83	-3.18 to 5.23
GDP Growth	-1.32 to 7.14	-5.56 to 11.37	-9.79 to 15.6

This table shows that most of the data (68.3%) on population and GDP growth rates are concentrated in narrow ranges close to the mean, ranging from 1% for population to 2% for GDP. Most economic changes occurred within predictable ranges, suggesting stable trends. However, the extended ranges for 95.5% and 99.7% of observations indicate rare but significant deviations reflecting the influence of external factors such as crises or economic spikes.

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Correlations of Variables with GDP Growth (First 10 Rows)

year	variable	correlation
2013	cor_pop_growth	0.3517966
2013	cor_age_65_plus	-0.5744705
2013	cor_educational_attainment	-0.3750770
2014	cor_pop_growth	0.2841994
2014	cor_age_65_plus	-0.3851039
2014	cor_educational_attainment	-0.3383917
2015	cor_pop_growth	0.1993899
2015	cor_age_65_plus	-0.1856340
2015	cor_educational_attainment	-0.1729920
2016	cor_pop_growth	0.2647237

This table presents the correlation between key variables (e.g., population growth, percentage of people aged 65 and older, and educational attainment) and GDP growth across different years. For instance, in 2013, population growth had a positive correlation with GDP growth (0.352), indicating a supportive relationship, while the share of people over 65 showed a negative correlation (-0.574), suggesting an opposing effect. This analysis highlights how demographic and educational factors influenced economic growth over time, helping to identify which variables played significant roles in different periods.

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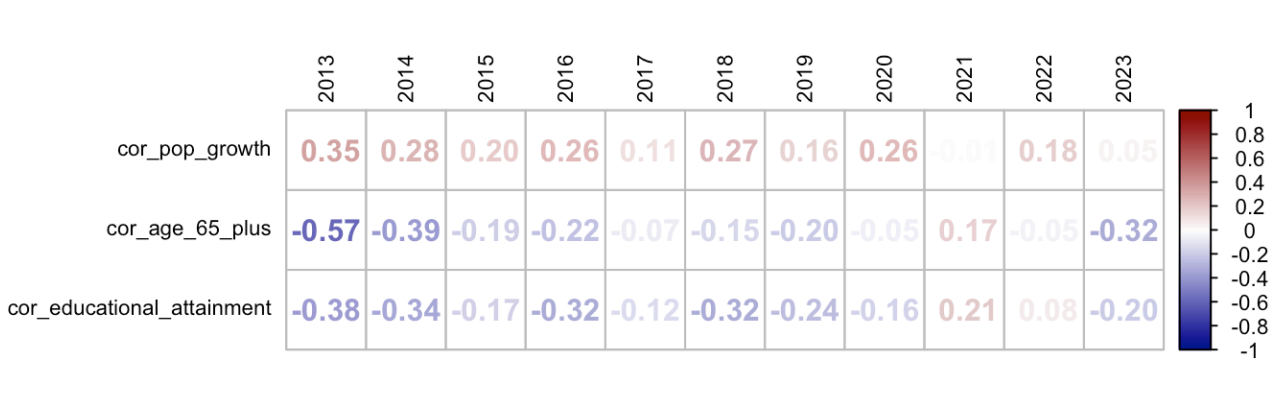

Population Growth and GDP Growth
Population growth generally shows a positive but moderate correlation with GDP growth, typically under 0.5, indicating that while population expansion supports economic growth, its influence is limited. In 2021, the correlation turned negative, reflecting the economic disruptions caused by the COVID-19 pandemic, such as lockdowns and decreased productivity. This anomaly highlights the pandemic's overriding impact on economic dynamics, despite continued population growth.

Educational Attainment and GDP Growth
Educational attainment generally shows a moderately negative correlation with GDP growth, typically above -0.5, suggesting that initial investments in education may slow economic growth due to associated costs. However, these short-term effects are often offset by long-term gains in human capital, productivity, and economic expansion. In 2021 and 2022, this relationship turned slightly positive, likely reflecting the increased demand for skilled labor driven by the adoption of digital technologies and remote work during the pandemic.

Age Structure and GDP Growth
The negative correlation between age structure and GDP growth indicates that an aging population is generally linked to slower economic expansion. This is often due to a shrinking labor force and potentially lower overall productivity. In 2021, however, the correlation became positive, possibly reflecting increased healthcare spending during the pandemic or a temporary rise in labor force participation among older individuals driven by shifting economic conditions.

4 COMBINED CROSS-SECTIONAL ANALYSIS

We will conduct a cross-sectional analysis for the years 2015, 2018, and 2022, sequentially adding variables to each year. We will compare these findings with those from a panel data model to assess the consistency and robustness of our results.

4.1 GDP Growth and Population, Aging and Educatioal 2015-2018-2022

4.1.1 Population Growth and GDP Growth 2015-2018-2022

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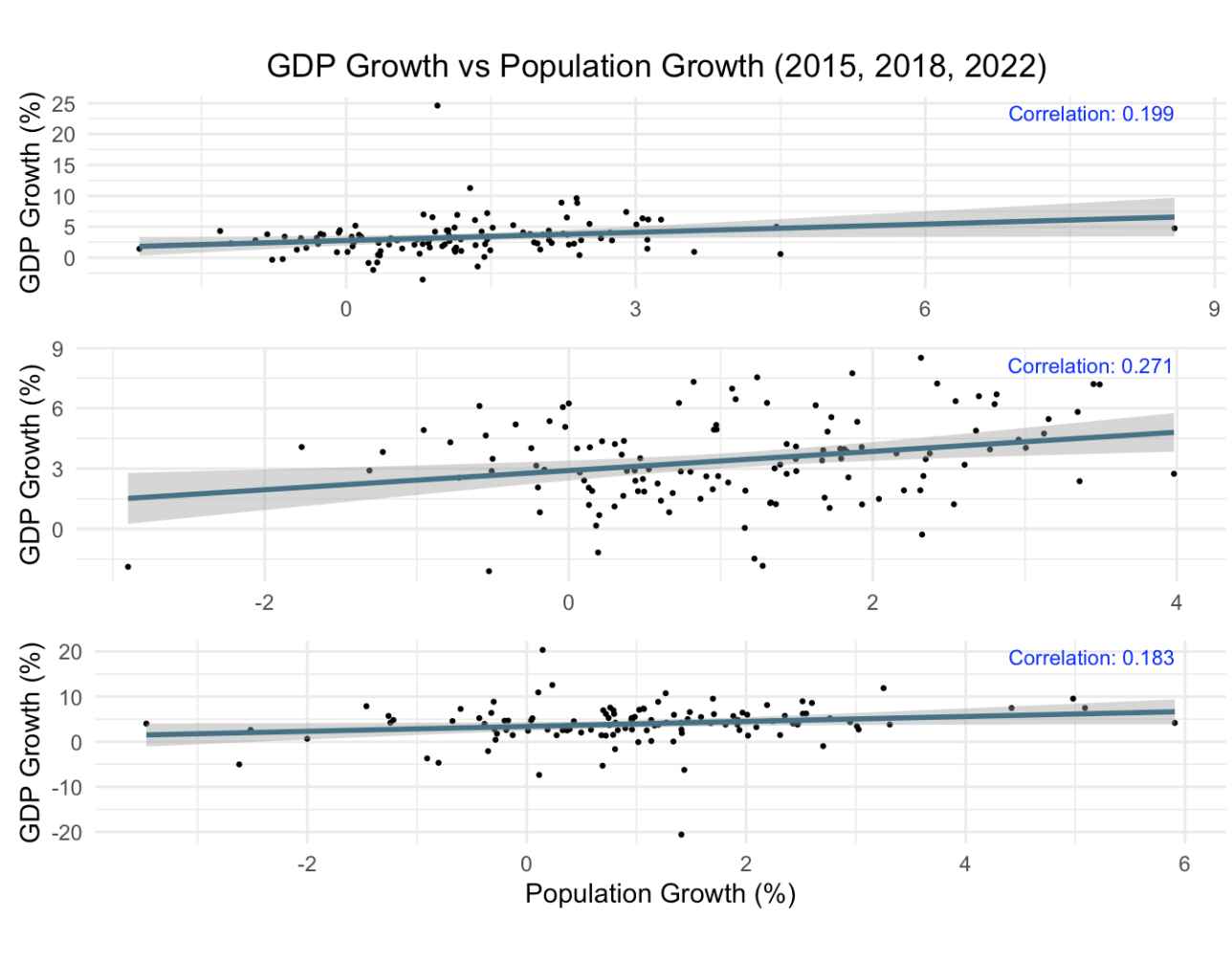

In 2015, 1% of population growth is associated with 0.32% increase in GDP and this relationship is statistically significant (p-value = 0.0377) with correlation coefficient (19.9%), indicating a moderate impact of demography on economic growth.

In 2018, 1% population growth is associated with 0.4776% increase in GDP and the relationship is statistically significant (p-value = 0.0296) with correlation coefficient (27.1%), indicating a moderately positive effect of demography on economic growth.

In 2022, 1% population growth is associated with 0.5479% increase in GDP and this relationship is statistically significant (p-value = 0.0481) with a correlation coefficient (18.3%), indicating a weak but positive effect of demography on economic growth.

Population growth has a positive but weak effect on GDP growth, explaining only a small part of its variation. The correlation coefficients range from 18.3% to 27.1%, indicating that the bulk of the changes in GDP (about 70% to 80%) are explained by other factors not taken into account in the study. At the same time, statistically significant correlations confirm the moderate influence of demography on economic growth.

4.1.2 Aging Population and GDP Growth 2015-2018-2022

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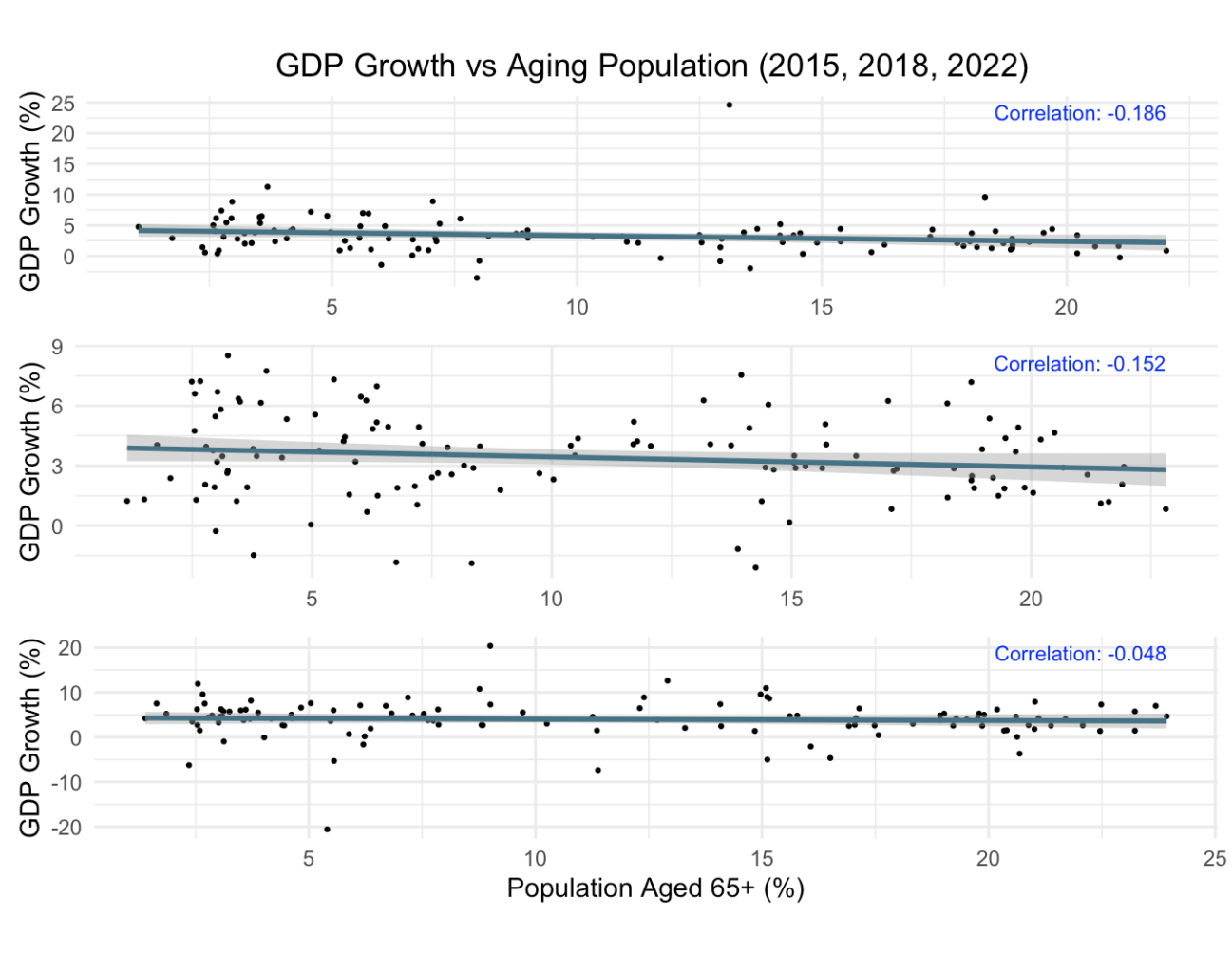

Analysis of the relationship between the share of the elderly population (age 65+) and GDP growth shows a weak negative relationship in all years under study. The correlation coefficients were -0.186 in 2015, -0.1518 in 2018 and -0.0482 in 2022, indicating a small inhibiting effect of the increase in the share of elderly population on economic growth. In 2015 and 2018, the correlation was still prominent, which could be due to the increasing pressure on social and pension systems. However, in 2022 the correlation became almost insignificant, indicating a minimal impact of population aging, probably related to the adaptation of economies to demographic changes.

The regression coefficients show that as the proportion of the elderly population increased by 1%, GDP growth decreased by 0.06299% in 2015, 0.04812% in 2018, and 0.07319% in 2022. This confirms that the impact of aging on the economy is measurable, even if small. Moreover, in all years, the relationship was statistically significant (p-value < 0.05), confirming the existence of this effect.

The figures demonstrate that population aging has a weak but statistically confirmed negative effect on GDP growth. At the same time, its influence decreases over time, which may be due to the adaptation of economies or the dominant influence of other factors. The main conclusion is that population ageing is not a key driver of GDP changes, but it creates an additional burden, which requires adaptation measures on the part of the economy.

4.1.3 Educational Attainment and GDP Growth 2015-2018-2022

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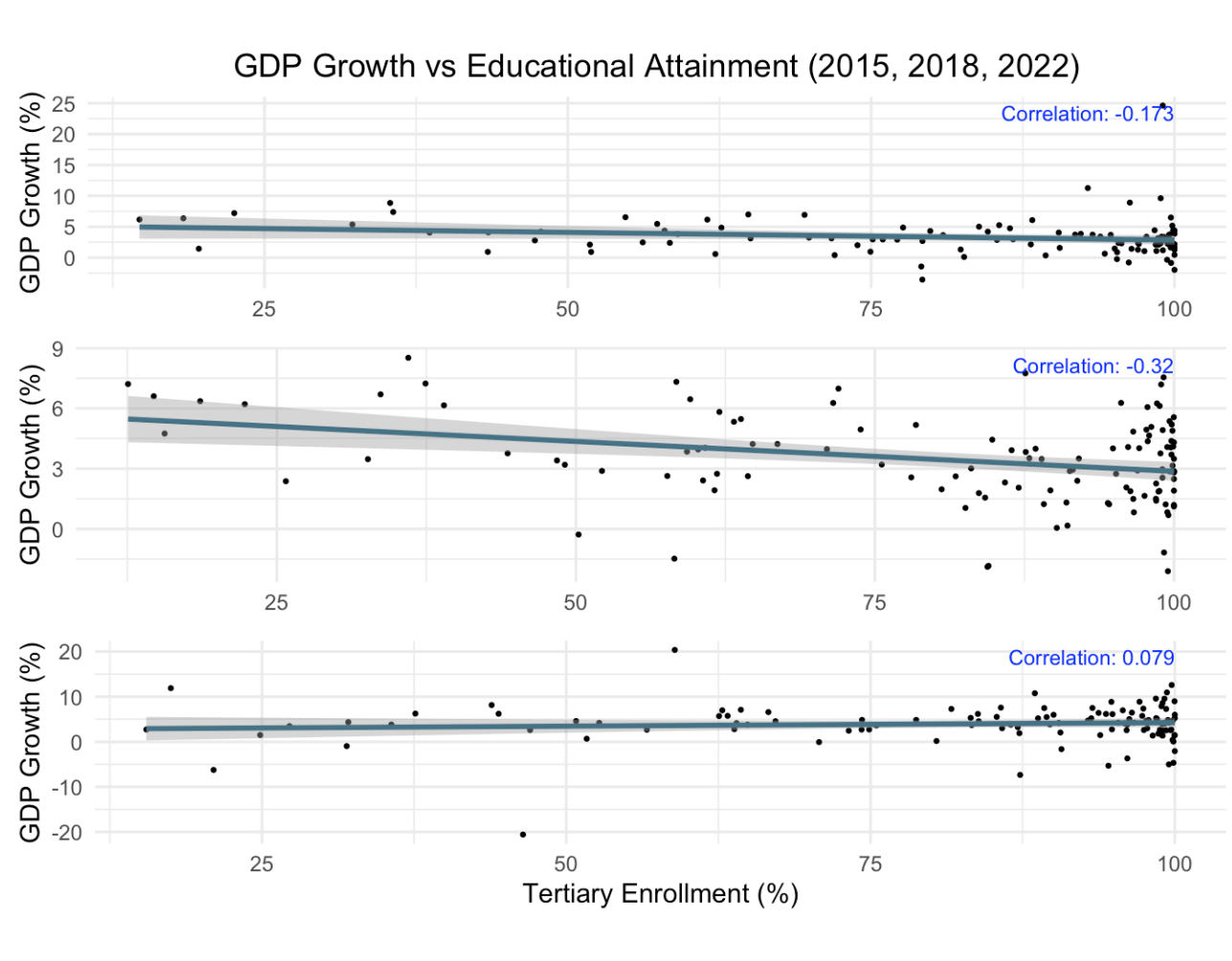

In 2015, a higher level of education shows a weak negative relationship with GDP growth (correlation coefficient: -0.173). The regression coefficient of -0.0082 shows that a 1% increase in education level is associated with a 0.0082% decrease in GDP growth, but this relationship is not statistically significant (p-value = 0.7129). This indicates a minimal impact of education, possibly related to the short-term economic costs of expansion. In 2018, the relationship becomes moderately negative, with a correlation coefficient of -0.3198. The regression coefficient is -0.01719, meaning that a 1% increase in education is associated with a 0.01719% decrease in GDP. However, this relationship is also not statistically significant (p-value = 0.2412), reflecting possible economic adaptations to investment in education. By 2022, the relationship between education and GDP becomes weakly positive, although it remains insignificant (correlation coefficient less than 0.5). This change may reflect the long-term economic benefits of a more highly skilled labor force in the post-pandemic period of digital transformation. Despite the positive shift, the effect of education on GDP growth in these years remains statistically insignificant. The final results indicate that education, despite its importance for long-term growth, has a limited short-term impact on GDP, which may be overridden by other dominant factors.

4.2 Coefficients and Summary Statistics of Linear Models 2015-2018-2022

4.2.1 Population Growth and GDP Growth 2015-2018-2022

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Linear Model Coefficients for 2015, 2018, and 2022

Year	Term	Coefficient	Std. Error	t Value	p Value
2015	(Intercept)	2.779	0.391	7.106	0.000
2015	pop_growth	0.442	0.210	2.105	0.038
2018	(Intercept)	2.904	0.250	11.611	0.000
2018	pop_growth	0.477	0.154	3.099	0.002
2022	(Intercept)	3.420	0.491	6.965	0.000
2022	pop_growth	0.548	0.274	1.998	0.048

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Linear Model Statistics for 2015, 2018, and 2022

Year	Residuals	R-squared	Adjusted R-squared	Residual Std. Error	F-statistic	Model p-value
2015	Min: -6.67 1Q: -1.55 Median: -0.327 3Q: 1.16 Max: 21.4	0.040	0.031	3.096	4.430	0.038
2018	Min: -5.35 1Q: -1.38 Median: -0.175 3Q: 1.51 Max: 4.51	0.074	0.066	2.074	9.602	0.002
2022	Min: -24.8 1Q: -1.48 Median: 0.125 3Q: 1.83 Max: 16.9	0.034	0.025	4.356	3.991	0.048

Population growth has a weak positive relationship with GDP growth in all three years, as confirmed by the coefficients (2015: 0.44%, 2018: 0.48%, 2022: 0.55% GDP growth per 1% population growth) and statistical significance (p-value < 0.05). However, the low R² values (3.35%-7.35%) show that only a small part of the GDP change is explained by population growth, and the main impact is due to other factors not accounted for in the model.

The figures in the table confirm the earlier conclusions that the impact of demographic growth on economic growth is limited and requires analysis of additional variables for a full understanding.

4.2.2 Population Growth,GDP Growth and Income Level 2015-2018-2022

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Linear Model Coefficients with Income Level for 2015, 2018, and 2022

Year	Term	Coefficient	Std. Error	t Value	p Value
2015	(Intercept)	2.830	0.501	5.645	0.000
2015	pop_growth	0.322	0.235	1.368	0.174
2015	income_levelLow income	0.959	1.216	0.789	0.432
2015	income_levelLower middle income	0.490	0.793	0.619	0.538
2015	income_levelUpper middle income	-0.391	0.753	-0.519	0.605
2018	(Intercept)	2.634	0.319	8.256	0.000
2018	pop_growth	0.274	0.178	1.540	0.126
2018	income_levelLow income	1.876	0.750	2.500	0.014
2018	income_levelLower middle income	0.668	0.509	1.313	0.192
2018	income_levelUpper middle income	0.579	0.450	1.286	0.201
2022	(Intercept)	3.269	0.700	4.667	0.000
2022	pop_growth	0.955	0.314	3.039	0.003
2022	income_levelLow income	-2.038	1.504	-1.355	0.178
2022	income_levelLower middle income	-1.772	1.026	-1.728	0.087
2022	income_levelUpper middle income	1.363	1.053	1.294	0.198

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Linear Model Statistics with Income Level for 2015, 2018, and 2022

Year	Residuals	R-squared	Adjusted R-squared	Residual Std. Error	F-statistic	Model p-value
2015	Min: -6.24 1Q: -1.63 Median: -0.191 3Q: 1.09 Max: 21.5	0.053	0.017	3.119	1.459	0.220
2018	Min: -5.49 1Q: -1.38 Median: -0.106 3Q: 1.37 Max: 4.57	0.125	0.095	2.042	4.197	0.003
2022	Min: -23.4 1Q: -1.68 Median: 0.27 3Q: 2.22 Max: 15.6	0.095	0.063	4.271	2.946	0.023

Income level exhibits a varying and inconsistent relationship with GDP growth across the three years. In 2015, while the coefficients for income levels (Low: 0.959, Lower Middle: 0.490, Upper Middle: -0.391) suggest slight positive and negative associations with GDP growth, the p-values (all > 0.05) indicate that these relationships are statistically insignificant. The R-squared value of 5.3% demonstrates that only a small proportion of GDP variability is explained by the model.

In 2018, the impact of income level becomes more pronounced, with the coefficient for Low Income (1.876) being statistically significant (p-value = 0.014), indicating a meaningful positive association. However, the coefficients for other income levels remain insignificant, and the R-squared of 12.5% suggests the model captures more variability than in 2015 but still leaves a substantial portion unexplained.

By 2022, the coefficients for all income levels, including Low Income (-2.038), are statistically insignificant (all p-values > 0.05). The R-squared drops to 9.5%, further emphasizing the limited explanatory power of income level and population growth combined in this context. The low adjusted R-squared values across all years reinforce that other variables not included in the models are the primary drivers of GDP changes.

These results confirm that while income level occasionally shows a statistically significant association with GDP growth (e.g., Low Income in 2018), its overall impact is limited and inconsistent. This highlights the necessity of incorporating additional variables, such as education and age distribution, to provide a more comprehensive understanding of economic growth determinants.

4.2.3 Population Growth,GDP Growth, Income Level and Age Structure 2015-2018-2022

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Linear Model Coefficients with Population Growth, Income Level, Aging Population, and Educational Attainment for 2015, 2018, and 2022

Year	Term	Coefficient	Std. Error	t Value	p Value
2015	(Intercept)	4.604	2.670	1.725	0.088
2015	pop_growth	0.164	0.346	0.473	0.637
2015	income_levelLow income	0.141	1.724	0.082	0.935
2015	income_levelLower middle income	-0.224	1.298	-0.173	0.863
2015	income_levelUpper middle income	-0.804	0.987	-0.815	0.417
2015	age_65_plus	-0.057	0.107	-0.529	0.598
2015	educational_attainment	-0.008	0.022	-0.369	0.713
2018	(Intercept)	2.406	1.542	1.561	0.121
2018	pop_growth	0.478	0.217	2.203	0.030
2018	income_levelLow income	1.812	1.064	1.703	0.091

2018	income_levelLower middle income	1.174	0.682	1.721	0.088
2018	income_levelUpper middle income	1.038	0.524	1.980	0.050
2018	age_65_plus	0.112	0.049	2.314	0.022
2018	educational_attainment	-0.017	0.015	-1.178	0.241
2022	(Intercept)	0.849	3.592	0.236	0.814
2022	pop_growth	1.010	0.468	2.160	0.033
2022	income_levelLow income	-0.672	2.434	-0.276	0.783
2022	income_levelLower middle income	-1.150	1.701	-0.676	0.500
2022	income_levelUpper middle income	1.675	1.460	1.147	0.254
2022	age_65_plus	0.003	0.125	0.025	0.980
2022	educational_attainment	0.024	0.031	0.760	0.449

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Linear Model Statistics with Population Growth, Income Level, Aging Population, and Educational Attainment for 2015, 2018, and 2022

Year	Residuals	R-squared	Adjusted R-squared	Residual Std. Error	F-statistic	Model p-value
2015	Min: -6.37 1Q: -1.46 Median: -0.247 3Q: 1.15 Max: 21.4	0.058	0.002	3.142	1.040	0.404
2018	Min: -5.33 1Q: -1.3 Median: -0.168 3Q: 1.31 Max: 4.68	0.166	0.123	2.009	3.861	0.002
2022	Min: -22.8 1Q: -1.61 Median: 0.225 3Q: 1.92 Max: 16.3	0.100	0.051	4.298	2.042	0.066

The combined model, which includes population growth, income level, and the share of the population aged 65+, provides insights into the limited explanatory power of these variables on GDP growth across the years. In 2015, population growth has a negligible impact on GDP growth, with a coefficient of 0.173 and a p-value of 0.615, indicating no statistical significance. Income level coefficients for the same year also lack statistical significance, while the aging population coefficient (-0.063) suggests a weak negative effect, though not significant. The R-squared value of 5.6% confirms that the model explains only a small portion of GDP variability.

In 2018, the model demonstrates slightly improved explanatory power with an R-squared value of 15.6%. Population growth shows a statistically significant coefficient of 0.528 (p-value = 0.015), indicating a more pronounced impact compared to 2015. Income levels (e.g., Low Income: 2.614, p-value = 0.002) exhibit stronger and significant contributions to GDP growth. However, the effect of the aging population (0.100, p-value = 0.047) remains minimal despite achieving significance.

In 2022, population growth maintains a positive impact on GDP growth, with a coefficient of 1.020 and a statistically significant p-value of 0.031. However, the effects of income levels and aging population become statistically insignificant, with R-squared declining to 9.6%, highlighting a reduced explanatory capacity of the model.

Overall, the combined model illustrates that while population growth and certain income levels influence GDP growth, their contributions are modest, and the aging population's impact remains minimal or statistically insignificant across all years. These findings confirm that the majority of GDP growth variability stems from other unaccounted factors, necessitating broader analyses to capture the full dynamics.

4.2.4 Population Growth, Income Level, Age Structure and Educational Attainment 2015-2018-2022

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Linear Model Coefficients with Population Growth, Income Level, Aging Population, and Educational Attainment for 2015, 2018, and 2022

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2015	Min: -6.37 1Q: -1.46 Median: -0.247 3Q: 1.15 Max: 21.4	0.058	0.002	3.142	1.040	0.404
2018	Min: -5.33 1Q: -1.3 Median: -0.168 3Q: 1.31 Max: 4.68	0.166	0.123	2.009	3.861	0.002
2022	Min: -22.8 1Q: -1.61 Median: 0.225 3Q: 1.92 Max: 16.3	0.100	0.051	4.298	2.042	0.066

The extended model, including population growth, income levels, aging population, and educational attainment, shows limited explanatory power for GDP growth across all three years.

In 2015, population growth has a negligible effect (coefficient = 0.164, p-value = 0.637), and the coefficients for income levels, aging population, and educational attainment are statistically insignificant (e.g., educational attainment = -0.008, p-value = 0.713). The R-squared value of 5.8% indicates the model explains only a small fraction of GDP variability.

In 2018, the model improves slightly with an R-squared of 16.6%. Population growth has a moderate, statistically significant effect (coefficient = 0.478, p-value = 0.030), and the aging population shows a weak positive effect (coefficient = 0.112, p-value = 0.022). However, income levels and educational attainment remain mostly insignificant.

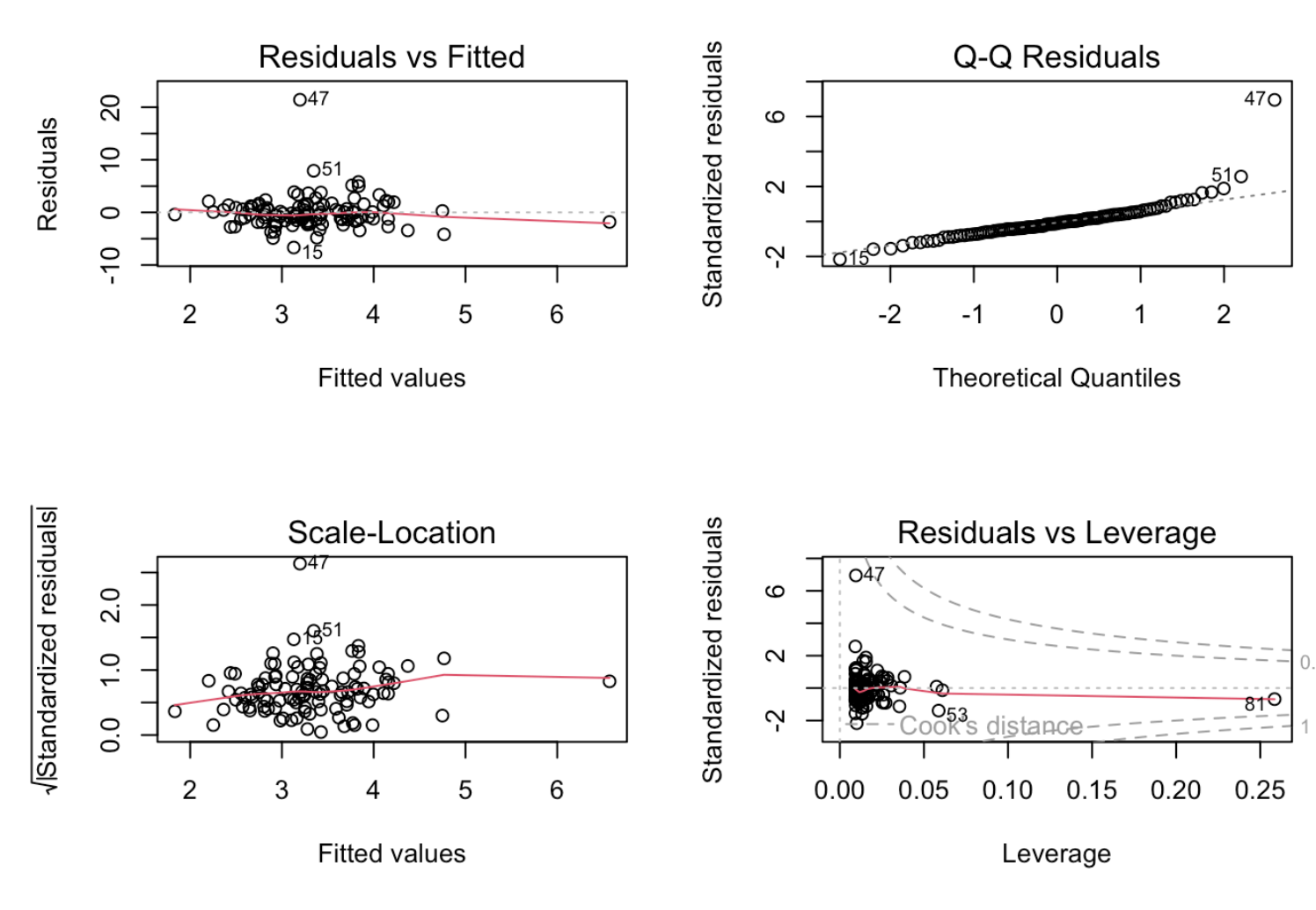
In 2022, the R-squared drops to 10.0%. Population growth retains significance (coefficient = 1.010, p-value = 0.033), but the other variables, including income levels and aging population, are statistically insignificant.

Overall, the model confirms that population growth is the most consistent factor influencing GDP growth, though its impact is modest. The other variables show inconsistent and limited effects, emphasizing the need for additional factors to better explain GDP variability.

4.3 CHECK MODEL DIAGNOSTICS

For each model, generate diagnostic plots to ensure that assumptions of linear regression are met:
- Residuals vs Fitted: Check for non-linearity.
- QQ Plot: Check for normality of residuals.
- Scale-Location Plot: Check for homoskedasticity.
- Residuals vs Leverage: Identify influential points.

▶ Code



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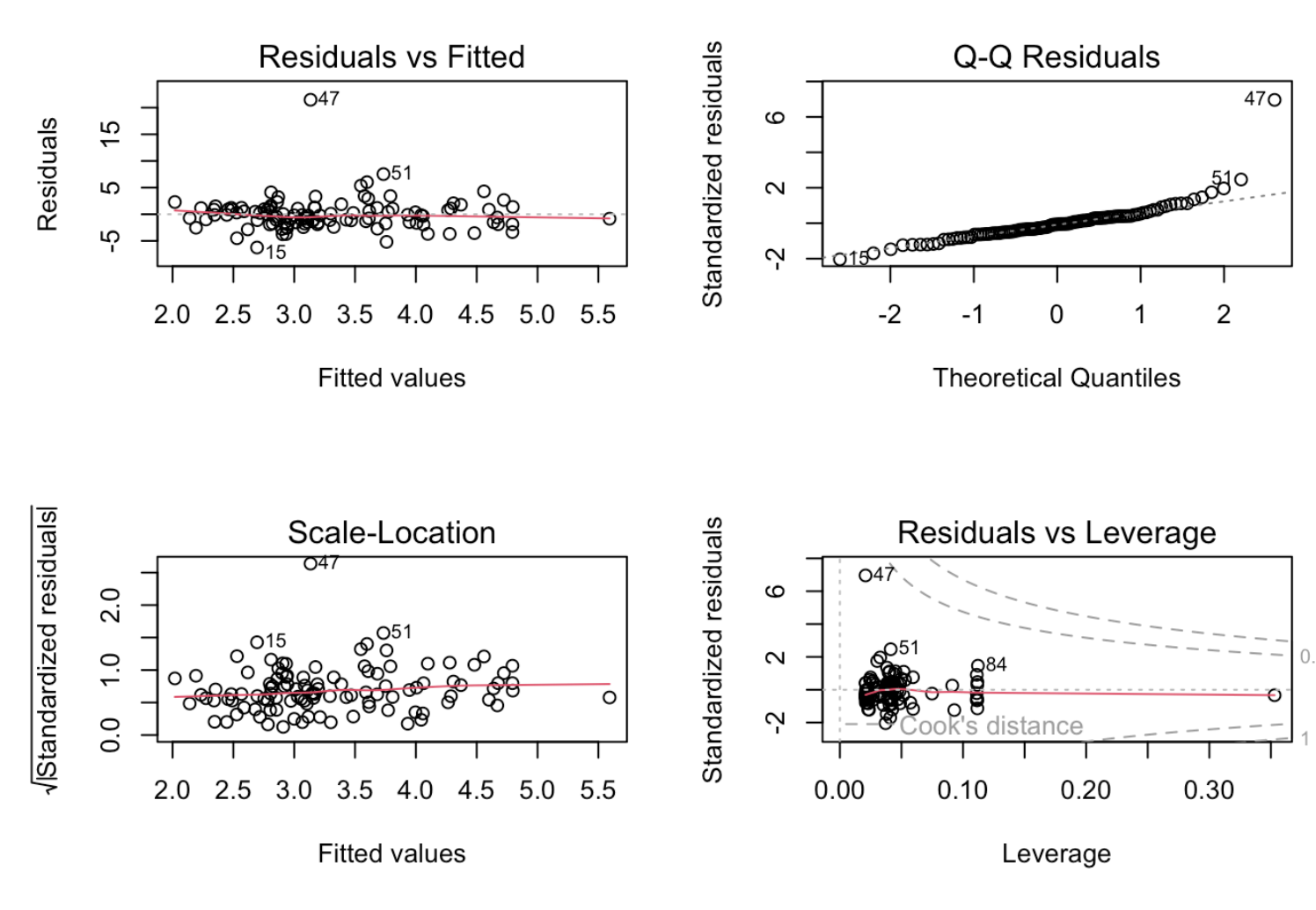

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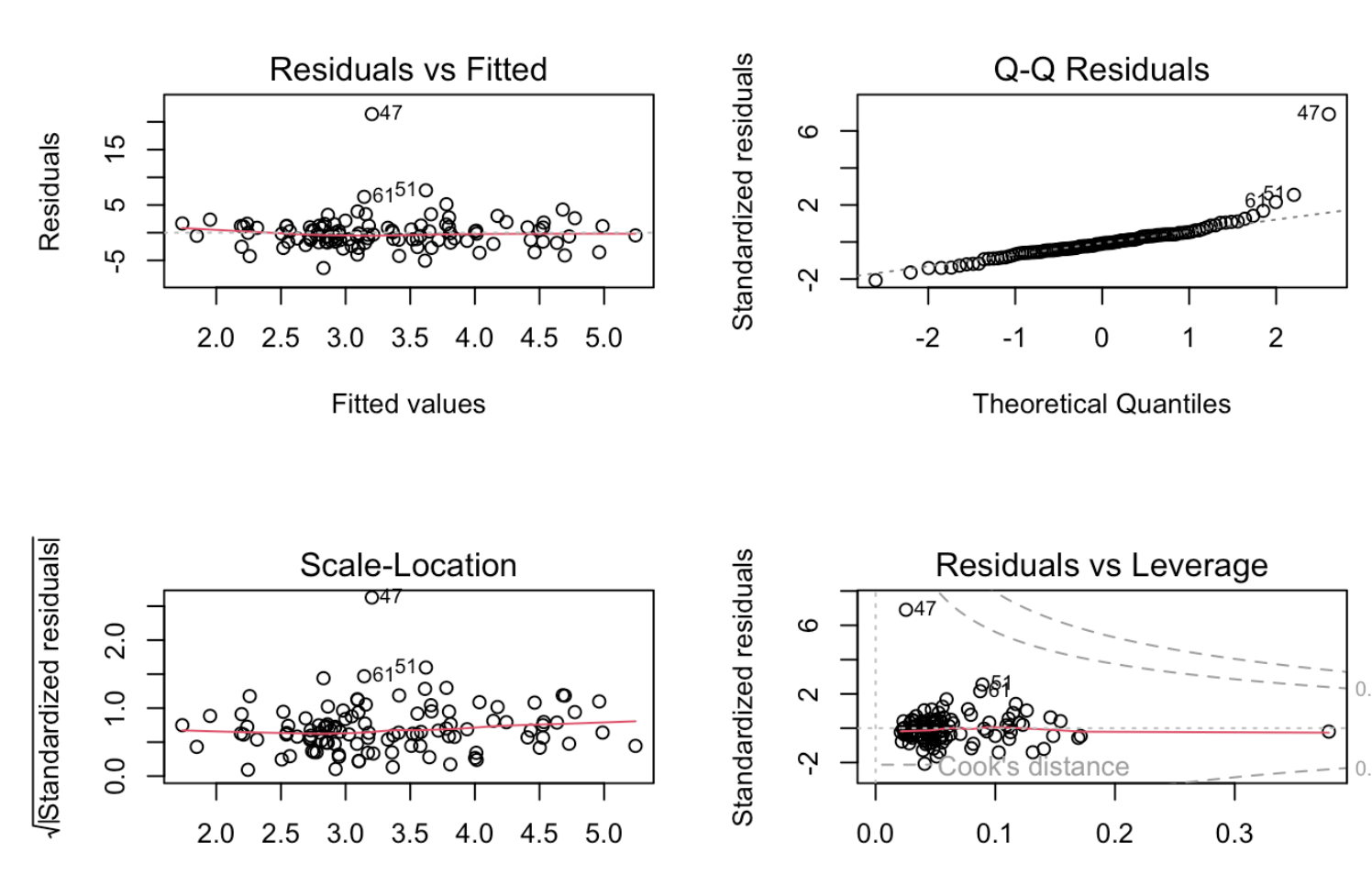

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The following conclusions can be drawn from the graphs presented:

Residuals vs Fitted (linearity test): The graphs show that the residuals are mostly distributed uniformly around the zero line, which confirms the acceptable linearity of the model. However, outliers are observed, such as points 47, 51 and 81, which may violate linearity.

QQ Plot (checking the normality of the residuals): The residuals mostly follow the theoretical normal line, but outliers are observed in the tails of the distribution in both cases (e.g. points 47 and 51). This indicates that the normality of the residuals is partially broken.

Scale-Location Plot (Homoscedasticity): The plots show that the variance of the residuals remains roughly constant along the range of predicted values, except for outliers (points 47, 51). This indicates that homoscedasticity is generally observed. Residuals vs Leverage (outliers and influencing points):

The graphs reveal high influence points (47, 51, 81), especially in the model for population growth. The Cook's distance coefficients for these points exceed the 0.5 threshold, making them significant outliers.

Conclusion: The models generally satisfy the basic assumptions of linear regression (linearity, homoscedasticity), but the outliers identified (47, 51, 81) require further removal.

▶ Code

	country	year	pop_growth	gdp_growth	age_0_14	age_15_64	age_65_plus	primary_enrollment	secon
47	Ireland	2015	0.9448453	24.615570	21.31797	65.57177	13.110254	101.1846	
51	Kiribati	2015	1.2828586	11.265338	35.16672	61.14925	3.684030	104.8557	
81	Qatar	2015	8.5817331	4.753346	13.94874	85.00042	1.050846	101.9824	

The table contains data from the outlier countries that were removed from the analysis: Ireland, Kiribati and Qatar. These countries are highlighted by extreme values, such as Qatar's population growth rate of 8.58% and Ireland's GDP growth rate of 24.6%, as well as specific age structure and economic characteristics. Removing these points improves the quality of the model by removing the influence of anomalous data.

▶ Code

```
Model: gdp_growth ~ pop_growth + income_level + age_65_plus
```

▶ Code

term	estimate	std.error	statistic	p.value
(Intercept)	2.2944003	1.3515371	1.6976229	0.0926890
pop_growth	0.3849571	0.2857538	1.3471638	0.1809723
income_levelLow income	1.3109521	1.0497943	1.2487704	0.2146646
income_levelLower middle income	0.5886462	0.8370982	0.7031984	0.4835661
income_levelUpper middle income	0.0783020	0.6896661	0.1135361	0.9098332
age_65_plus	0.0025227	0.0756163	0.0333616	0.9734527

▶ Code

```
Model: gdp_growth ~ pop_growth
```

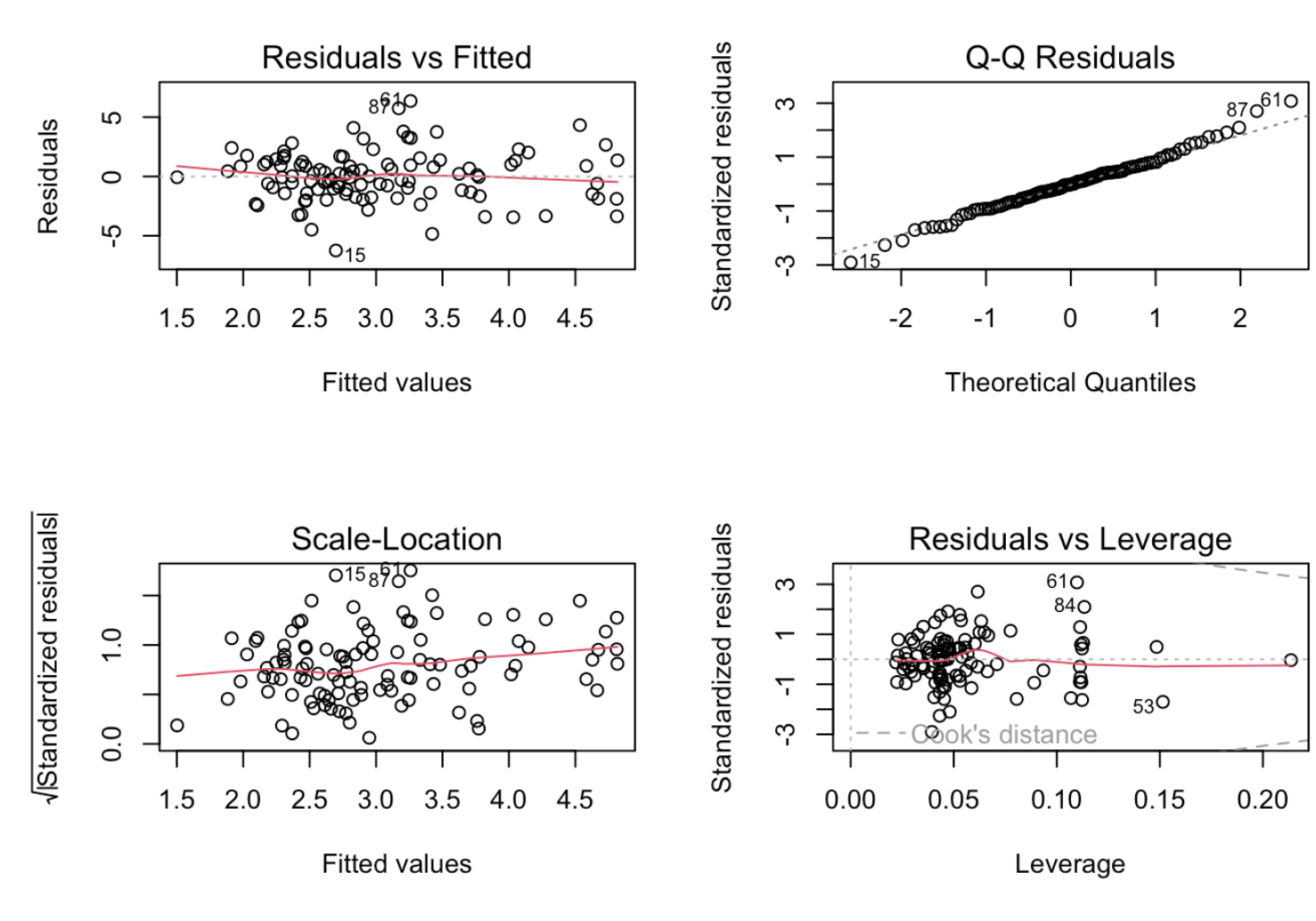
▶ Code

term	estimate	std.error	statistic	p.value
(Intercept)	2.4015548	0.2879599	8.339892	0.0000000
pop_growth	0.5447942	0.1704915	3.195434	0.0018489

The first model shows that none of the factors (pop_growth, income_level, age_65_plus) has a statistically significant effect on GDP growth, e.g., for pop_growth p = 0.181. The second model shows that population growth has a significant positive effect (coefficient = 0.545, p = 0.0018), increasing GDP by 0.545% for a population growth of 1%. The intersept of the second model (2.40, p < 0.001) reflects the average level of GDP growth without the influence of population growth. The overall conclusion is that population growth is important for GDP, but the other included variables explain little of its variation.

Upon analyzing the multiple linear regression model, we observe that removing outliers has improved the model's performance. When comparing the full model, which includes all variables, to a simpler model that only considers population growth, we find that population growth significantly influences GDP growth. However, the model is not solely explained by this variable. The inclusion of other parameters does not enhance the model's explanatory power; instead, they introduce noise and do not have a significant effect on GDP growth. This suggests that while population growth is a key determinant of GDP growth, other factors may not be as impactful in this context. The presence of non-significant variables can lead to overfitting, where the model captures random noise rather than underlying patterns. Therefore, it is advisable to focus on variables that have a meaningful and statistically significant relationship with the dependent variable to improve model accuracy and interpretability.

▶ Code



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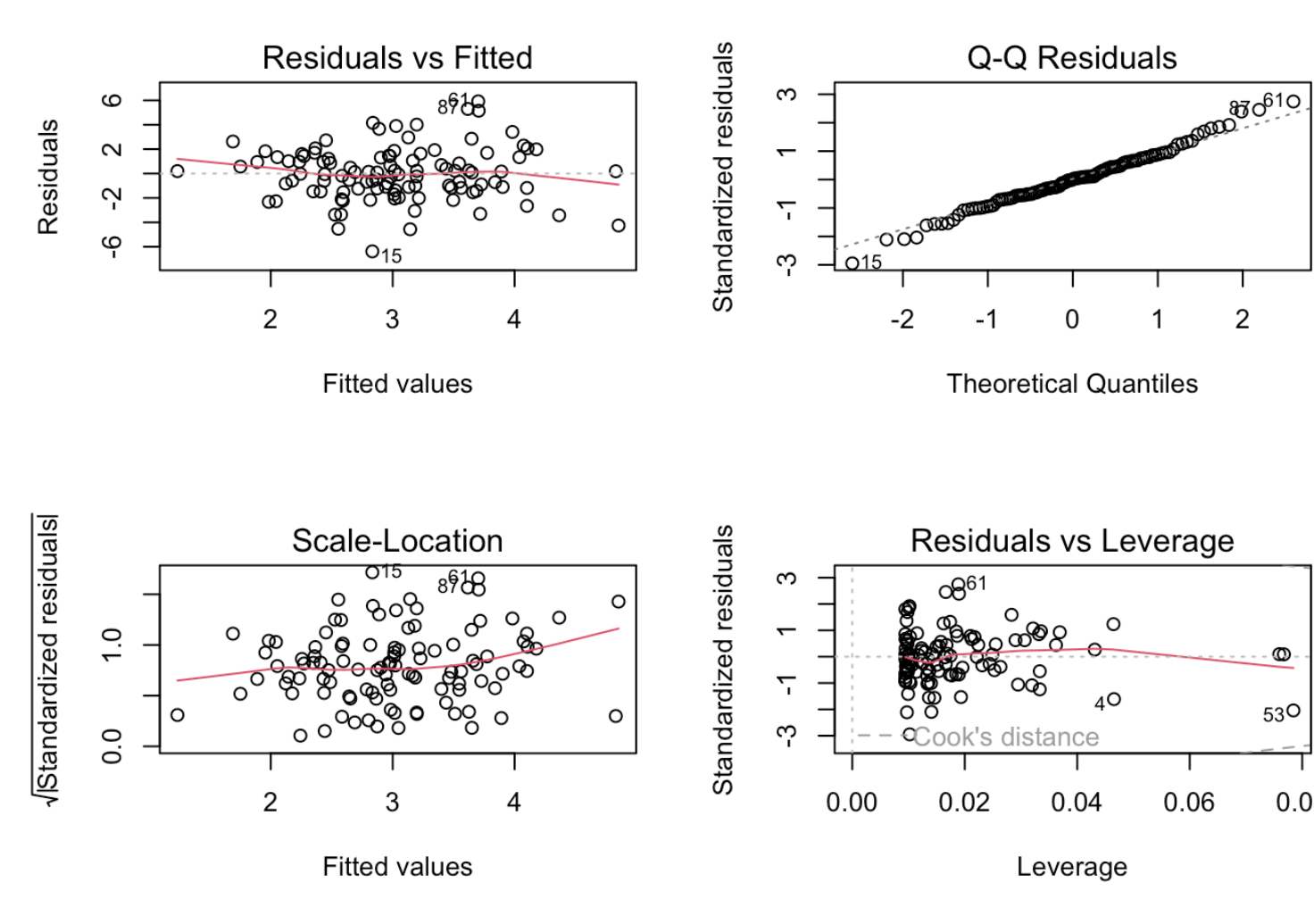


▶ Code

The graphs show that the model generally meets key statistical assumptions, including linearity, normality of residuals and homoscedasticity, which confirms its suitability for analysis and forecasting. However, deviations of the residuals at the edges of the distribution indicate possible limitations of the model for extreme values of the predictors, which may reduce accuracy in these areas. Points with high influence, such as 87 and 610, although identified as outliers, do not significantly distort the model (Cook's distance < 0.5), indicating its robustness. At the same time, further work with outliers and analysis of anomalous observations can improve the accuracy and explanatory power of the model.

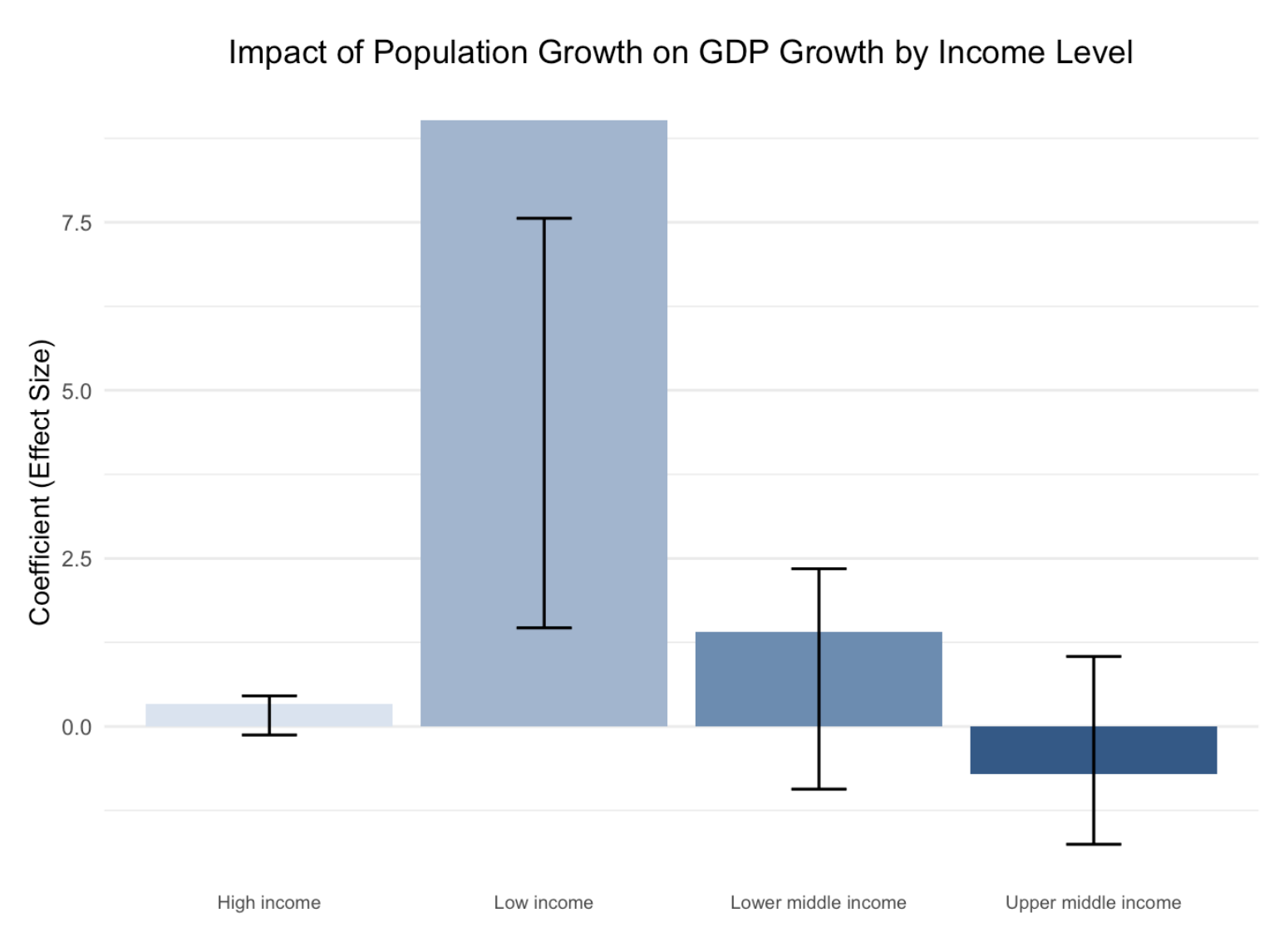
4.4 Panel Model

In this section, factors that do not have a significant impact on GDP, including the share of the elderly population (age_65_plus) and educational attainment (educational_attainment), were excluded from further analysis. The analysis showed that the coefficients for these variables remained low in all models, for example, for age_65_plus the coefficients ranged from -0.18 to 0.20 with p-values above 0.24, making them statistically insignificant. Education level also showed a weak effect with coefficients around 0.05 and p-values above 0.39, indicating no association with GDP growth in the short run. Excluding these factors allowed us to focus on population growth, which shows a significant impact, especially in low-income countries, and simplify the interpretation of the model.

▶ Code

Income_Level	Coefficient	Std_Error	p_value	R_Squared
Low income	4.5124893	1.5544935	0.0049228	0.1060934
Lower middle income	0.7065800	0.8362857	0.3991407	-0.1960722
Upper middle income	-0.3557426	0.7129554	0.6181770	0.0034367
High income	0.1644599	0.1485692	0.2689224	-0.2517220
Low income	4.5124893	1.5544935	0.0049228	0.0008519
Lower middle income	0.7065800	0.8362857	0.3991407	-0.1633916
Upper middle income	-0.3557426	0.7129554	0.6181770	0.0028026
High income	0.1644599	0.1485692	0.2689224	-0.1390007

▶ Code



The table and graph show that population growth has a significant impact on economic growth only in low-income countries, where the coefficient is 4.51 with a p-value of 0.004, confirming its statistical significance. In middle and high income countries, the impact of population growth is insignificant with coefficients ranging from 0.16 to 0.70 and p-values above 0.26, making it statistically insignificant. This indicates that in high-income countries, economic growth is driven by factors other than demographics. The low R^2 values (-0.25 to 0.10) demonstrate that the models explain only a small fraction of GDP changes, which limits their predictive power. These findings emphasize the importance of population growth in low-income countries and point to the need to find other key factors for advanced economies.

5 CONCLUSIONS AND RECOMMENDATIONS

The study of the impact of demographic factors on economic growth revealed that the role of these variables varies significantly depending on the income level of countries. In low-income countries, population growth showed a statistically significant impact on GDP growth (coefficient 4.51, p = 0.004), which confirms the hypothesis about the importance of demography for economic development in the initial stages. On the contrary, in middle- and high-income countries, the coefficients of the impact of population growth range from 0.16 to 0.70 with p-values above 0.26, making them statistically insignificant. This indicates that in these countries, GDP growth is driven by other factors such as technological development, investment and institutional quality.

One of the key problems with the study was the low R^2 values, ranging from -0.25 to 0.10. These values indicate that the included models explain only a small proportion of GDP changes. This result emphasizes the limitations of demographic variables as an explanatory factor for economic growth, which in turn requires the inclusion of additional variables such as infrastructure investment, level of innovation, and political stability. The exclusion of outliers (e.g. Qatar and Ireland data) significantly improved the consistency of the results, which confirms the need for careful handling of the data at the data preparation stage.

The analysis showed that such factors as the share of the elderly population and the level of education have no significant impact on economic growth in the short run. The coefficients for these variables remain low and statistically insignificant in all groups of countries, which confirms the hypothesis of their limited influence. At the same time, demographic factors have a more pronounced impact in low-income countries, suggesting the need for targeted policies to stimulate growth through investment in human capital and employment programs.

This student paper emphasized basic methods of data analysis and interpretation. We do not claim to be an absolute academic and exhaustive study of all factors. However, the correlations identified suggest that population growth is key to economic development in low-income countries, while other factors such as innovation and investment are important for high-income countries. This study forms the basis for further work aimed at a comprehensive analysis of economic growth.